NovaTech Industries — Checkpoint 2.7

Supplier Performance & Compliance Data Generation

Project: Intelligent Procurement & Logistics Management Platform

Status: Implementation exercise

Version: Synthetic dataset v2

Reporting date: 2026-10-21 (fixed for reproducibility)

Seed: 2028

1. Objective

Extend the six validated reference and procurement datasets with historical delivery, quality, and supplier-category compliance evidence. This checkpoint **generates evidence only**: it does not approve quotations, compute final weighted supplier scores, or authorize purchase orders.

2. Input and output contracts

The generator reads these existing files from `data/synthetic/v2/`:

Input	Required columns
suppliers.csv	supplier_id, status
products.csv	product_id, category
supplier_products.csv	supplier_id, product_id

It writes three new files to the **same** directory without modifying the six existing datasets:

Output	Fields	Expected rows in current v2
supplier_deliveries.csv	delivery_id, supplier_id, planned_delivery_date, actual_delivery_date, on_time, status	177
supplier_quality_inspections.csv	inspection_id, supplier_id, inspection_date, inspected_units, accepted_units, rejected_units	66
supplier_compliance.csv	supplier_id, category, approval_status, valid_until	26

Compliance has one row per **supplier/category combination** present in `supplier_products.csv`, not one row per supplier.

3. Core business concepts

Reliability

Reliability measures the percentage of completed deliveries arriving on or before their promised date:

```
reliability_score = 100 * on_time_completed_deliveries / completed_deliveries
```

The generator stores both planned and actual dates, plus a derived `on_time` flag. Future scoring should recompute this flag from dates and check consistency.

Quality

Quality measures the accepted proportion of inspected units:

`quality_score = 100 * sum(accepted_units) / sum(inspected_units)`

For every inspection, `accepted_units + rejected_units = inspected_units` and all quantities must be nonnegative.

Compliance

A supplier's approval is **category-specific**. A quotation for a product category requires a matching supplier/category compliance record that is `APPROVED` and unexpired on the evaluation date. A supplier's `ACTIVE` status alone is not sufficient.

Insufficient history

NovaTech's confirmed policy `DP-006` is to flag insufficient evidence as `REVIEW_REQUIRED` for a Procurement Manager, not to invent quality or reliability scores. For later implementation, proposed (not yet formally approved) thresholds are **100 inspected units** and **10 completed deliveries** within the trailing 12 months. The generator intentionally gives `SUP012` only three deliveries and one small inspection to test this policy.

4. Implementation

Save the accompanying Python file at:

`src/data_generation/generate_supplier_performance_v2.py`

From the project root, run:

```
python3 src/data_generation/generate_supplier_performance_v2.py
```

Expected output with the existing 12-supplier v2 dataset:

```
Created supplier_deliveries.csv: 177 records
Created supplier_quality_inspections.csv: 66 records
Created supplier_compliance.csv: 26 records
Validation passed: identifiers, foreign keys, dates, quantities, and compliance categories.
Seed: 2028; reporting date: 2026-10-21; intentional low-history supplier: SUP012
Note: performance and compliance evidence is generated; no supplier score or PO approval is issued.
```

5. Practical validation

Check the files:

```
ls -lh data/synthetic/v2/supplier_*.csv
head -n 4 data/synthetic/v2/supplier_deliveries.csv
head -n 4 data/synthetic/v2/supplier_quality_inspections.csv
head -n 4 data/synthetic/v2/supplier_compliance.csv
```

Run the generator a second time: because the random seed and reporting date are fixed, the output should be identical.

Exercise A: Explain why `supplier_compliance.csv` has more than 12 rows.

Exercise B: Find `SUP012` in deliveries and inspections. Why must it be manually reviewed?

Exercise C: Explain why a supplier can be `ACTIVE` yet not eligible for a particular quotation.

Exercise D: Identify the validation that checks `accepted + rejected = inspected`.

6. Quality and limitations

- All generated records are fictional and not evidence of real supplier performance.

- Dates fall in a fixed 12-month reporting window ending 2026-10-21.

- Compliance edge cases include `PENDING` and `EXPIRED` statuses.

- The `on_time` field is derived from dates, not independently sampled.

- The generator validates identifiers, relationships, date ranges, and inspection totals before writing outputs.

- It does not create procurement-specific delivery evidence, exchange-rate evidence, or an audit-ready approval workflow; those belong in later checkpoints.

- The thresholds above are recommendations awaiting a policy decision, not an approved business rule.

7. Next checkpoint

Checkpoint 2.8 — Supplier Performance Metrics & Eligibility Integration: aggregate trailing-12-month quality/reliability metrics, enforce manual-review flags, and join compliance status with quotation eligibility. Only after validation will we implement the full 40/25/20/15 weighted recommendation model.